Mini Project 2 native batch result figures

This appendix charts actual saved official AI Studio batch results. It contains derived figures, not AI Studio GUI screenshots, and does not complete the 14 pending GUI photographs.

All nine algorithms share identical 18,000 training and 12,000 testing ID/label pairs. Positive class Hospitalized: 2,289 positive and 9,711 negative test rows. Modeling sample: 30,000; full prepared source: 1,048,575. Native metrics, fitted artifacts and source hashes were verified live before authoring.

Model	Accuracy	Precision	Recall	F1	AUC
Naive Bayes	0.8751	0.6910	0.6243	0.6560	0.8809
Generalized Linear Model	0.9032	0.8275	0.6225	0.7105	0.9015
Logistic Regression	0.8798	0.8731	0.4329	0.5789	0.9055
Fast Large Margin	0.8549	0.7433	0.3657	0.4902	0.7215
Deep Learning	0.9045	0.7746	0.7042	0.7378	0.9112
Decision Tree	0.9028	0.8292	0.6173	0.7077	0.7997
Random Forest	0.8909	0.8249	0.5435	0.6553	0.9051
Gradient Boosted Trees	0.8982	0.7743	0.6579	0.7114	0.9046
Support Vector Machine	0.9015	0.8449	0.5924	0.6965	0.8713

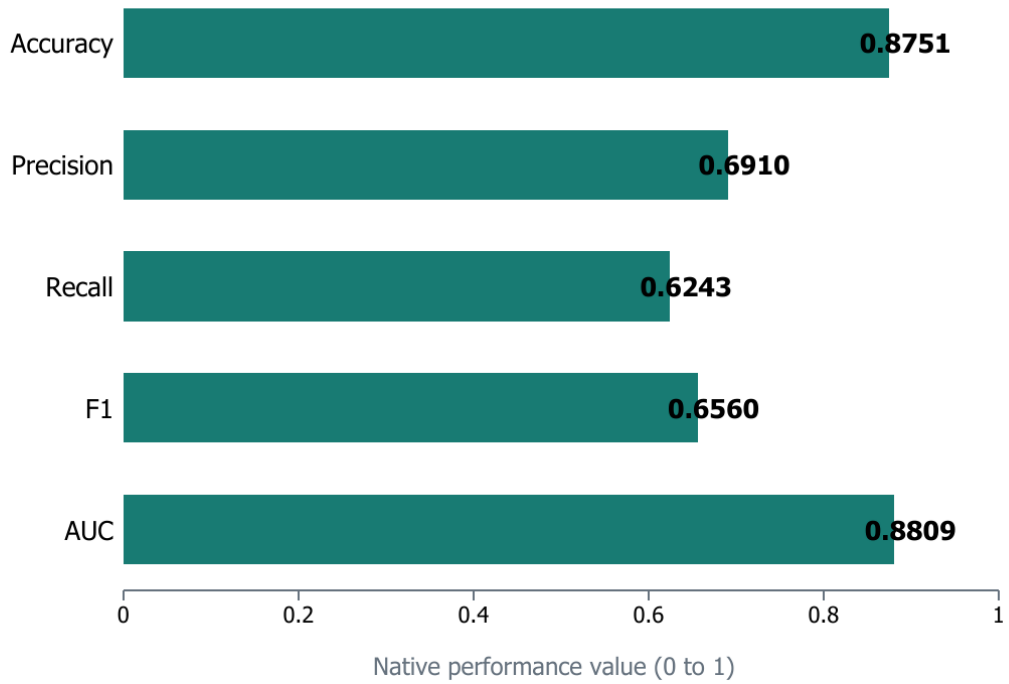
Deep Learning is selected in the educational common-test comparison. A single test comparison is not independent post-selection validation or proof of clinical deployment suitability. No repeated-run SD or confidence interval is inferred. AUC uses native confidence scores, not class labels.

Sources: canonical native_model_comparison.csv; common_test metric/prediction exports; same-fit native LR/DT/RF artifacts and RF learner weights in common_test/explanation_run. Exact values and per-figure source SHA256 are in figure_sources_and_values.json.

Naive Bayes

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	1,429
False positives	639
False negatives	860
True negatives	9,072

Hospitalized: 2,289
Outpatient: 9,711

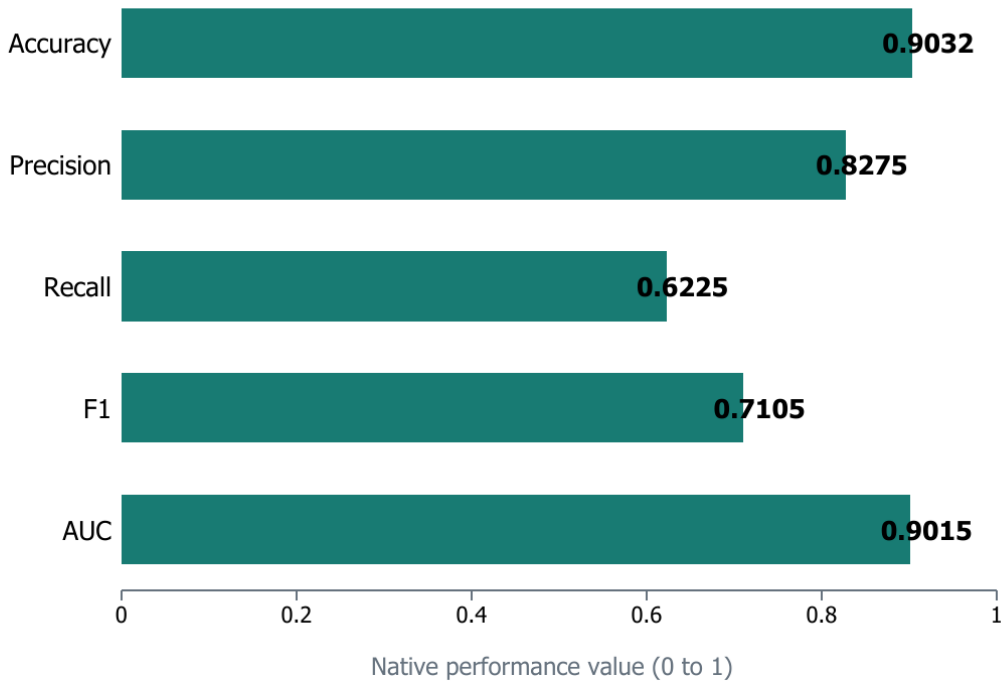
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Generalized Linear Model

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	**1,425**
False positives	**297**
False negatives	**864**
True negatives	**9,414**

Hospitalized: 2,289
Outpatient: 9,711

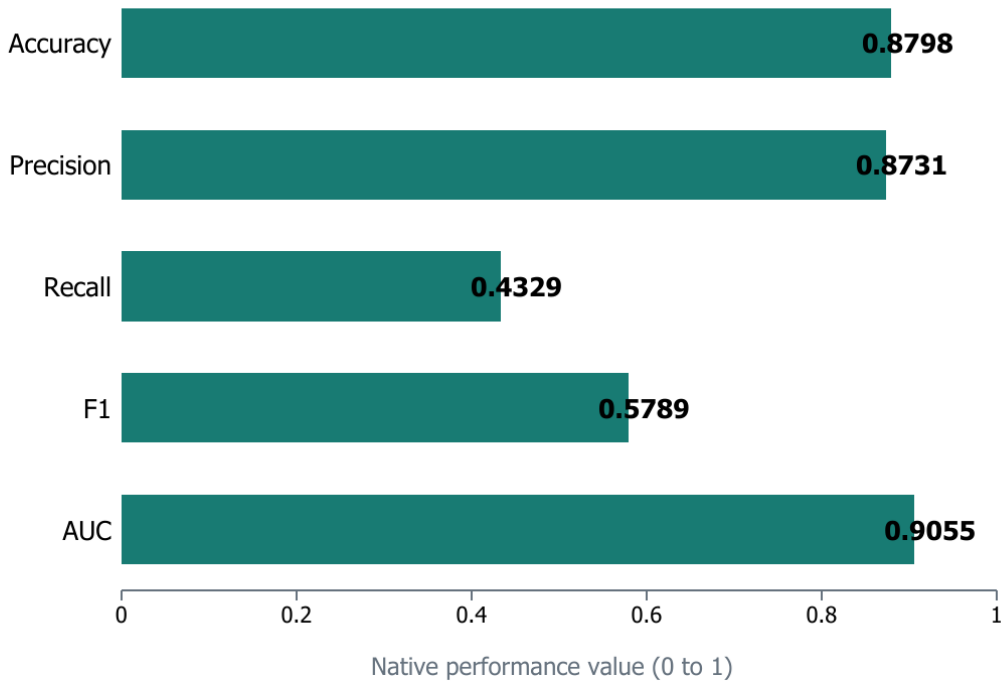
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Logistic Regression

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	**991**
False positives	**144**
False negatives	**1,298**
True negatives	**9,567**

Hospitalized: 2,289
Outpatient: 9,711

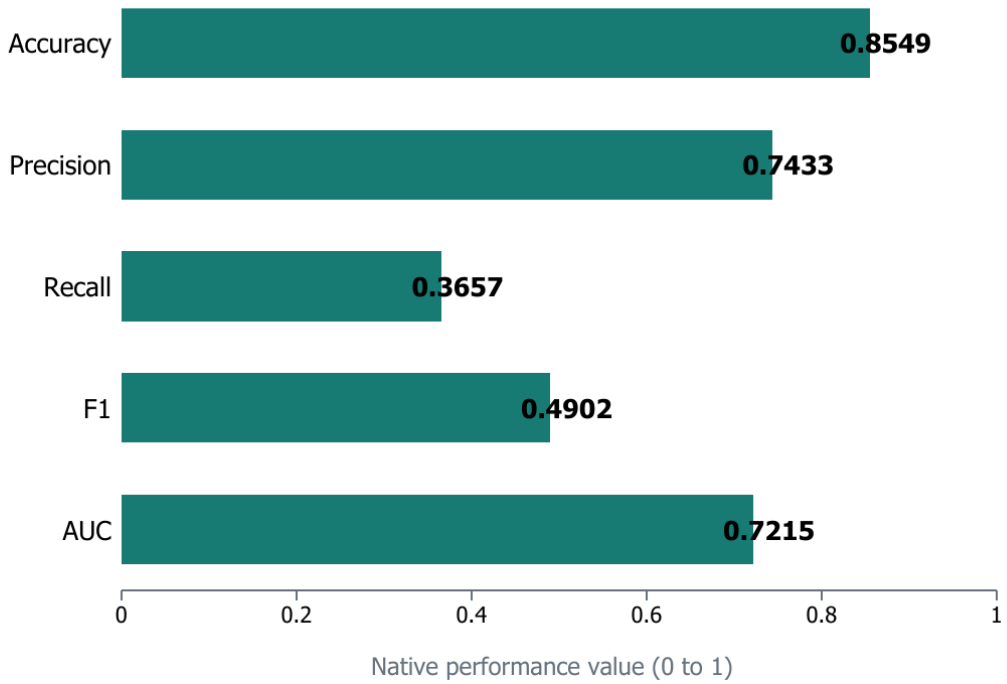
Chart of saved native exports; NOT an AI Studio GUI screenshot.

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Fast Large Margin

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	**837**
False positives	**289**
False negatives	**1,452**
True negatives	**9,422**

Hospitalized: 2,289
Outpatient: 9,711

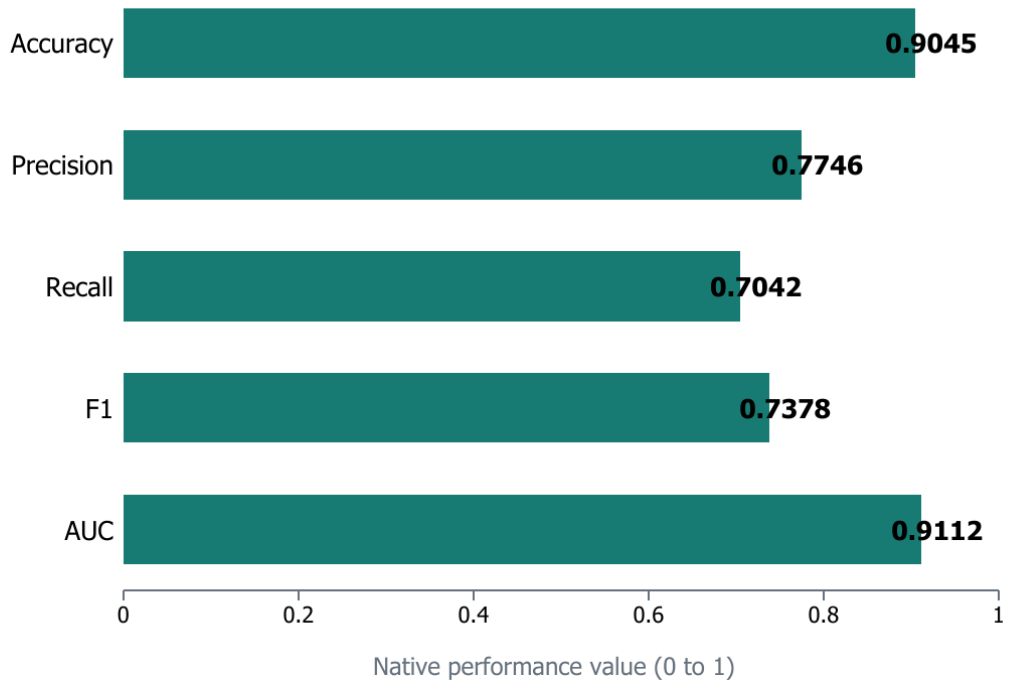
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Deep Learning

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	1,612
False positives	469
False negatives	677
True negatives	9,242

Hospitalized: 2,289
Outpatient: 9,711

Selected in this comparison

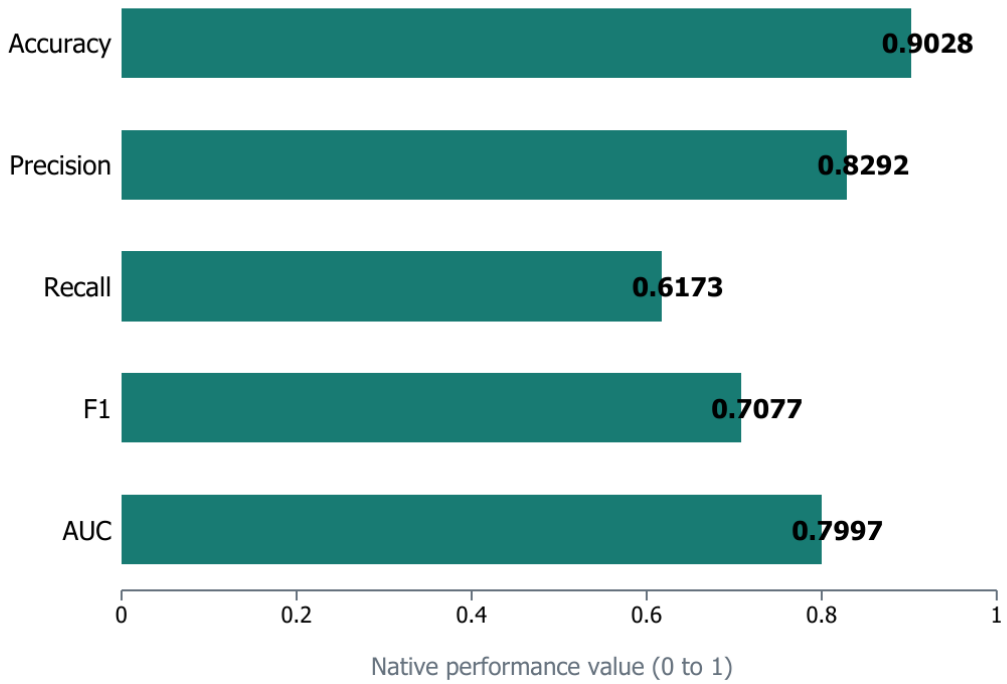
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Decision Tree

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	**1,413**
False positives	**291**
False negatives	**876**
True negatives	**9,420**

Hospitalized: 2,289
Outpatient: 9,711

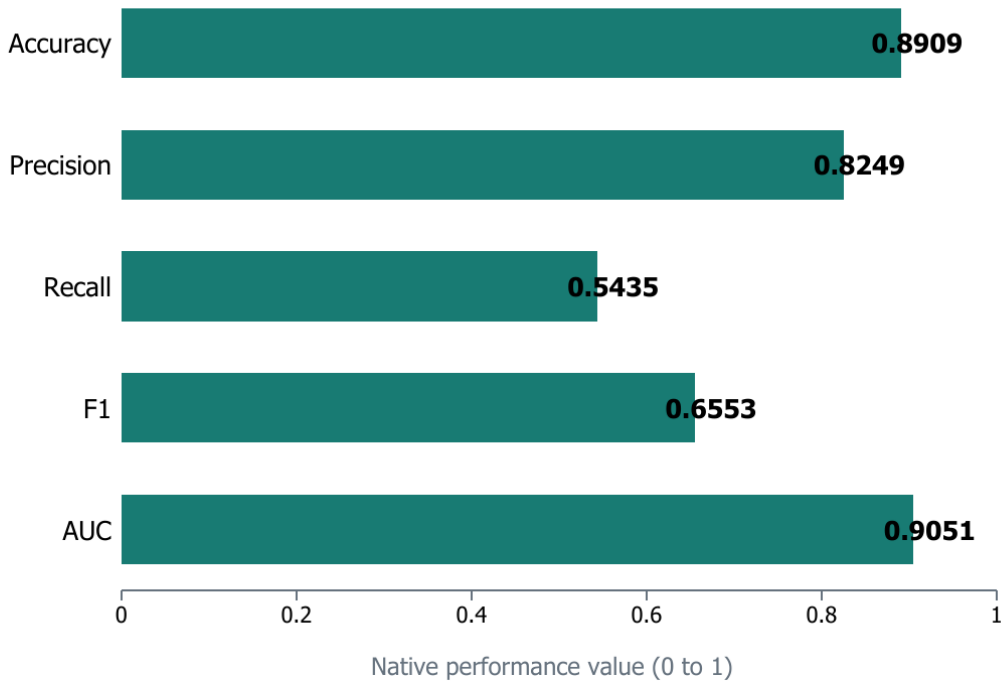
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Random Forest

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	1,244
False positives	264
False negatives	1,045
True negatives	9,447

Hospitalized: 2,289
Outpatient: 9,711

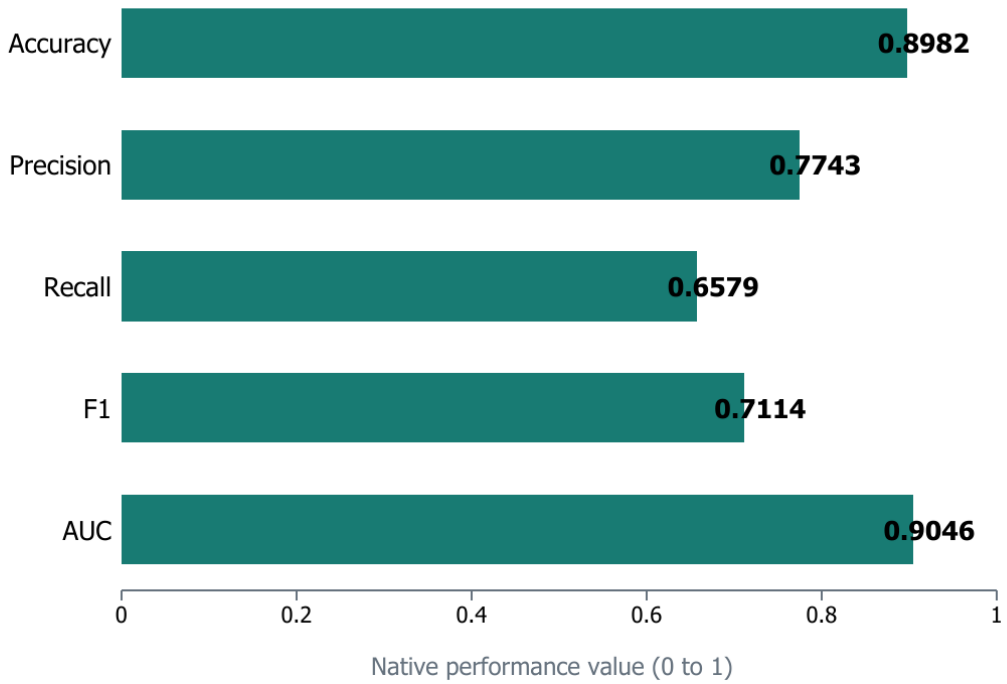
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Gradient Boosted Trees

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	1,506
False positives	439
False negatives	783
True negatives	9,272

Hospitalized: 2,289
Outpatient: 9,711

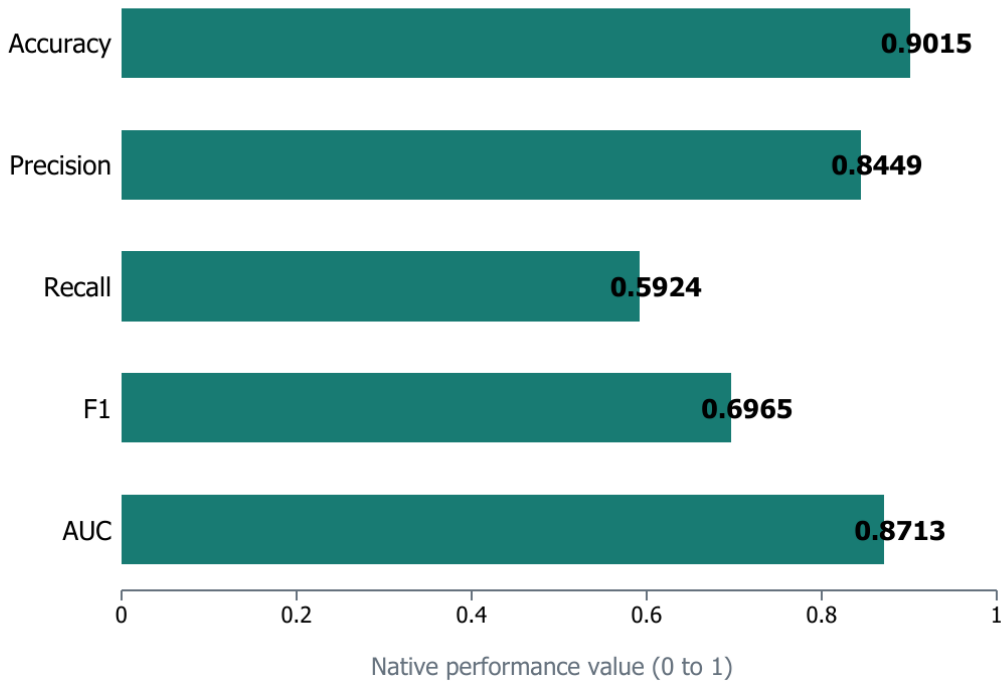
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Support Vector Machine

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Five metrics on the same full holdout; values are fractions from 0 to 1.



Actual native test counts

True positives	1,356
False positives	249
False negatives	933
True negatives	9,462

Hospitalized: 2,289
Outpatient: 9,711

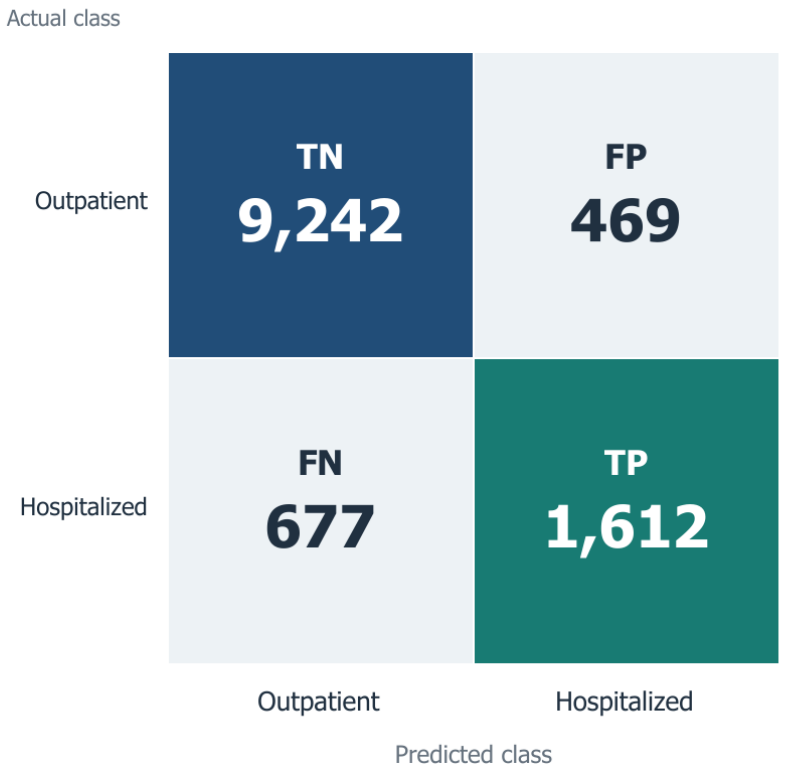
Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Deep Learning confusion matrix

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Rows are ACTUAL classes; columns are PREDICTED classes. Hospitalized confidence threshold: 0.50.



Verified 12,000-row holdout

Accuracy: 0.9045

Precision: 0.7746

Recall: 0.7042

F1: 0.7378

AUC: 0.9112

False negatives: 677 of 2,289 actual Hospitalized records. The four cells sum to 12,000.

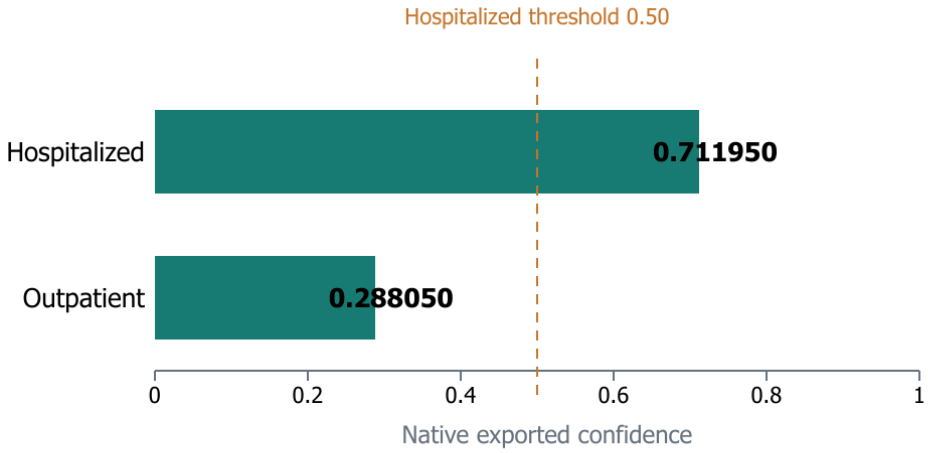
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Deep Learning native scoring example

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Audit row 348: actual Hospitalized; predicted Hospitalized. Audit ID is excluded from inputs.



0.711950 >= 0.50 => Hospitalized

Actual exported inputs

USMER: 2
MEDICAL_UNIT: 3
SEX: 1
PNEUMONIA: 1
AGE: 72
DIABETES: 2
COPD: 2
ASTHMA: 2
INMSUPR: 2
HIPERTENSION: 2
OTHER_DISEASE: 2
CARDIOVASCULAR: 2
OBESITY: 1
RENAL_CHRONIC: 2
TOBACCO: 2

Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Random Forest native feature importance

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Genuine learner weights output | criterion: gain_ratio | 15 normalized weights sum to 1.

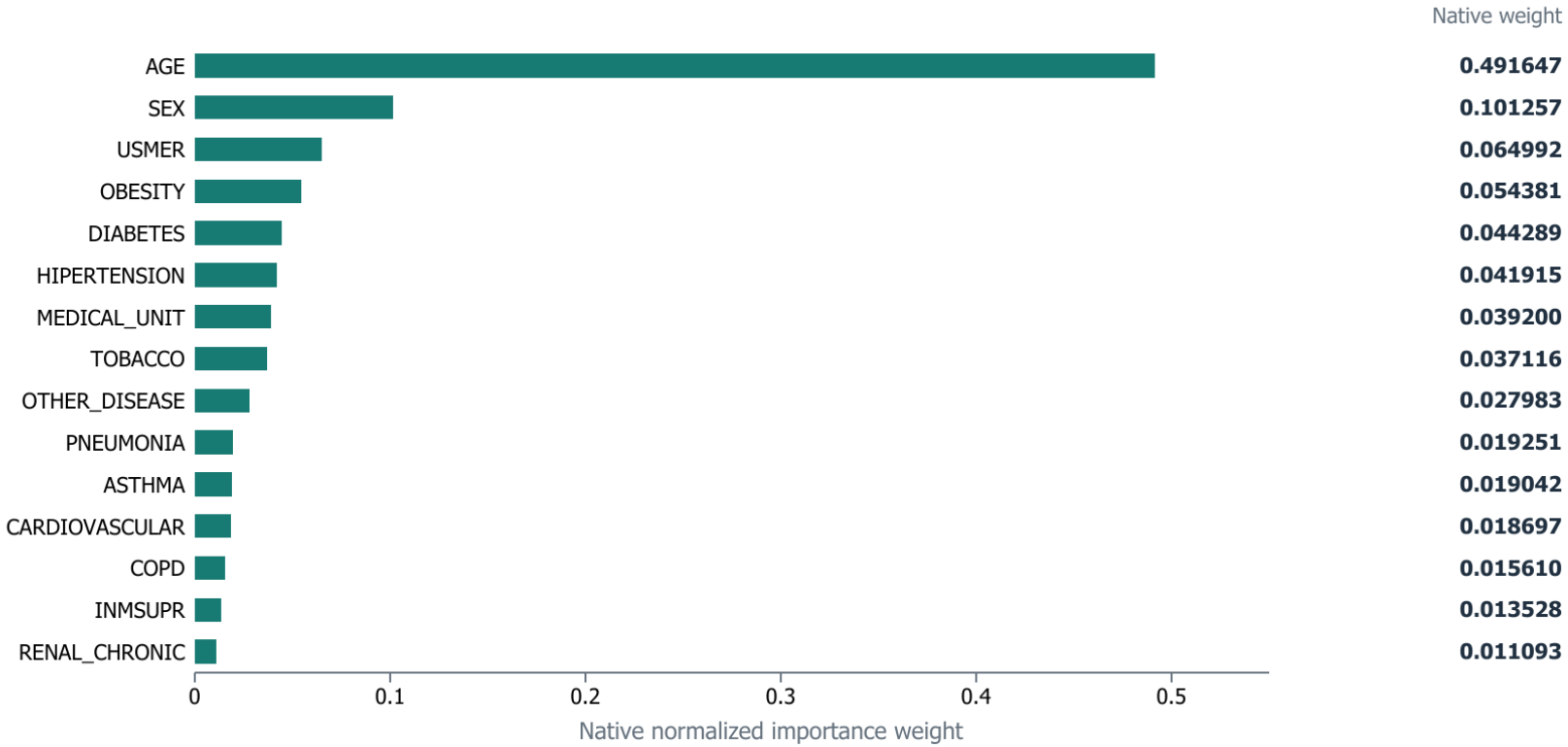


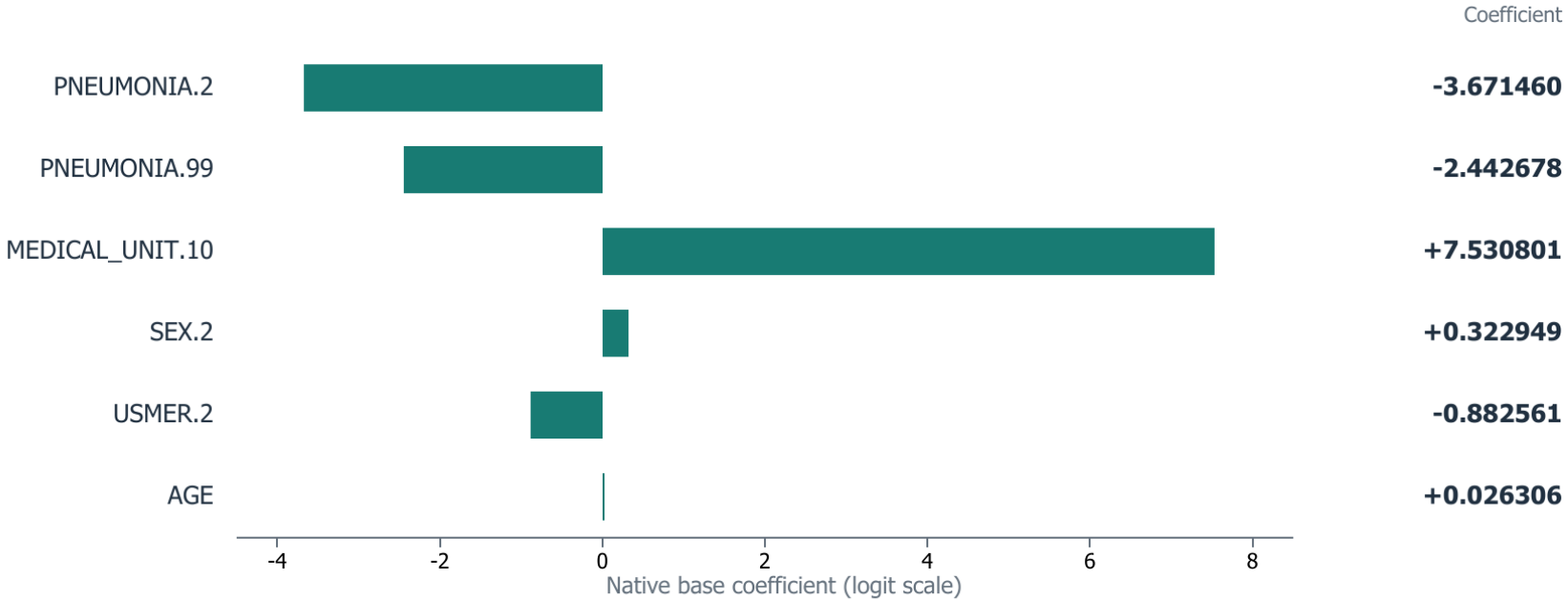
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Logistic Regression native coefficients

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

Evaluated common fit: base logit for Hospitalized. Final scores also use native calibration/cost wrappers.



Categorical level 1 is the omitted reference (inferred from native encoded names); AGE is per year.
Coefficient size depends on units/coding. Signs are not evidence of a causal effect.

Chart of saved native exports; NOT an AI Studio GUI screenshot.

Pipeline train: 18,000 | Modeling sample: 30,000 | Prepared source: 1,048,575 | No new model run

Decision Tree native root-to-leaf path

OFFICIAL AI STUDIO BATCH RESULTS | Common test: 12,000 | Positive: Hospitalized

One actual path; other branches are not shown. Leaf counts are training-node counts, not test counts.

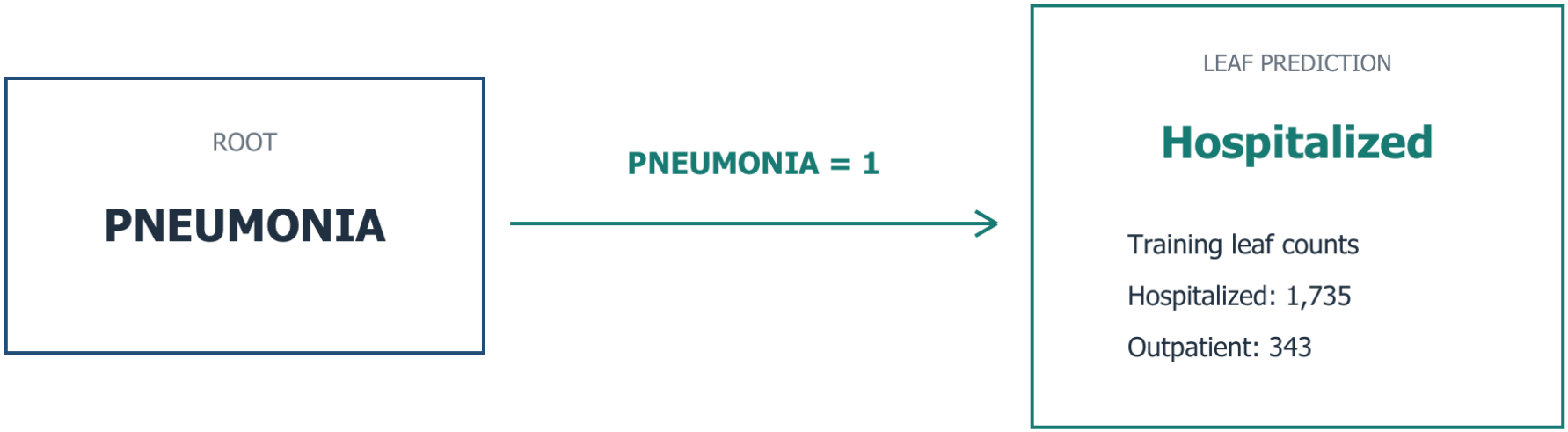


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